From Bag-of-Words to Large Language Models: A Unified Comparative System for Text Classification

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Abstract

This proposal presents **ClassiScope**, a unified comparative system for text classification that integrates four model families—sparse lexical features, static word embeddings, fine-tuned pretrained transformers, and prompting of instruction-tuned large language models (LLMs)—under a single data protocol, evaluation suite, and web interface. The project is motivated by an observation from our course exercises: on news topic classification, a simple word-frequency baseline slightly outperformed a fine-tuned BERT under identical splits, raising the question of *when* model sophistication actually pays off. We plan to answer this question with controlled experiments on two public benchmarks, an interpretability module that compares token-level explanations across model families, and robustness and efficiency analyses. The deliverables are an open-source system with a live demonstration, a controlled comparison study, and an ACL-format final report.

1 Team Members and Responsibilities

This project is completed individually. Jiahao Wu (Student ID: 2113454) is solely responsible for all components: problem formulation and related-work survey, system design and implementation, data preparation, all experiments and analysis, the final report and presentation, and the runnable demonstration.

2 Background and Motivation

Text classification is the canonical task of NLP, and its solutions have evolved through four technical generations: (i) sparse lexical features with linear classifiers, such as bag-of-words or TF–IDF with logistic regression; (ii) distributed word embeddings (Mikolov et al., 2013; Pennington et al., 2014) compressed into document vectors and fed to lightweight classifiers; (iii) fine-tuned pretrained transformers (Vaswani et al., 2017; Devlin et al., 2019); and (iv) instruction-tuned LLMs applied through zero-/few-shot prompting, without any task-specific training (Brown et al., 2020; Ouyang et al., 2022; Qwen Team, 2024).

These families embody different trade-offs among labeled-data requirements, compute cost, interpretability, and deployment effort, yet published comparisons are often fragmentary: datasets, splits, tuning budgets, and metrics differ from study to study. Our own course exercise made the issue concrete. On the NYT news topic classification task, with identical 80/10/10 stratified splits for every model, a plain word-frequency baseline reached slightly *higher* accuracy (99.1%) than a fine-tuned BERT (98.4%). Sophistication, in other words, must be earned rather than assumed. This observation motivates the central question of the project—*under what conditions does each model family actually pay off?*—and the system contribution: a single, controlled, and demonstrable platform in which all families are trained, explained, stress-tested, and compared under exactly the same protocol.

3 Problem Definition

We will build **ClassiScope**, a unified text-classification system, and use it to answer four research questions:

- **RQ1 (Accuracy).** How do the four model families compare on the same datasets, with identical splits, preprocessing, and metrics?
- **RQ2 (Robustness).** How do their predictions degrade under character-level perturbations (typos, spacing, casing) and cross-task transfer?
- **RQ3 (Interpretability).** What token-level evidence does each family rely on, and to what extent do explanations agree across families?

- **RQ4 (Efficiency).** What accuracy–cost trade-offs (trainable parameters, training time, inference latency) arise for practical deployment?

4 Proposed System and Implementation Plan

System functions. Given an input text, **ClassiScope** produces side-by-side predictions from all four model routes with confidence scores, token-level explanation highlights, and model metadata; given a configured dataset, it trains or loads all routes, runs the robustness and efficiency suites, and renders a comparison dashboard. A web interface (Gradio) exposes the full pipeline as a live demonstration.

Architecture. The system has three layers. (1) A *data layer* with unified loaders and preprocessing; datasets are shuffled and split once (80/10/10, stratified, fixed seed) and all models consume the same cached splits. (2) A *model layer* in which four routes implement a common `predict_with_explanation` interface: TF–IDF + logistic regression; Word2Vec/GloVe mean-pooled document vectors + logistic regression; fine-tuned BERT-base; and an instruction-tuned LLM (Qwen2.5, accessed via API, with a small local variant as fallback) under zero-/few-shot prompting. (3) An *analysis layer* providing interpretability (logistic-regression coefficients; gradient-based saliency for BERT; LLM rationales), the robustness suite, an efficiency profiler, and the web demo.

Implementation plan. Existing course-exercise code for the first three routes will be ported into the framework and refactored behind the common interface; the LLM route, explanation module, and web demo are built new. All experiments are scripted, seeded, and reproducible; the source code will ship with an English README covering the environment, dependencies, data acquisition, and run steps.

5 Evaluation Plan

We evaluate on two public benchmarks spanning two classification tasks: the NYT news topic dataset used in course exercises (Sandhaus, 2008) and the SST-2 sentiment dataset (Socher et al., 2013). All routes report **Accuracy** and **Macro-F1** on the same held-out test set, with paired bootstrap tests for significance. Robustness is measured as accuracy deltas under the perturbation suite and as cross-task transfer gaps. Efficiency is reported as trainable parameters, wall-clock training time, and per-sample inference latency. Interpretability is assessed qualitatively and quantitatively, by the overlap of top-attributed tokens across families. The four routes serve as baselines for one another; the study itself is the comparison, so no external system is required for a valid result.

6 Expected Outcomes

The project will deliver: (1) an open-source, reproducible system with a runnable demo (live demo or video, per course requirements); (2) a controlled comparison study answering RQ1–RQ4; (3) an eight-page ACL-format final report; and (4) final slides and a contribution statement. We hypothesize that (H1) the fine-tuned transformer is strongest in-domain when labeled data is plentiful, with sparse and embedding-based baselines close behind on easy topic tasks, as our course results suggest; (H2) LLM prompting is competitive without any task-specific training, especially in low-data regimes, but at higher latency and cost; and (H3) the families make complementary errors, which their respective explanations expose. The system is designed so that whichever hypotheses the data overturns, the resulting analysis remains a substantive contribution.

7 Project Timeline

Phase / Period	Tasks
Phase 1: Proposal	This document (due Oct. 12, 2026).
Weeks 1–2 (Oct. 13–26)	Unified data pipeline; port sparse, embedding, and BERT routes onto both datasets; demo skeleton.
Phase 2: Midterm (date TBD)	All four routes running with preliminary results; midterm report and progress slides.
Weeks 3–5 (Oct. 27–Nov. 16)	LLM prompting route; interpretability, robustness, and efficiency studies; full analysis.
Weeks 6–7 (Nov. 17–30)	Final report writing; demo polish and demo video; final slides and contribution statement.
Phase 3: Final (dates TBD)	Final report, slides, source code, demo, and contribution statement submitted; in-class presentation.

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